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Review

Timing is everything: advances in quantifying splicing kinetics

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Splicing is a highly regulated process critical for proper pre-mRNA maturation and the maintenance of a healthy cellular environment. Splicing events are impacted by ongoing transcription, neighboring splicing events, and *cis* and *trans* regulatory factors on the respective pre-mRNA transcript. Within this complex regulatory environment, splicing kinetics have the potential to influence splicing outcomes but have historically been challenging to study *in vivo.* In this review, we highlight recent technological advancements that have enabled measurements of global splicing kinetics and of the variability of splicing kinetics at single introns. We demonstrate how identifying features that are correlated with splicing kinetics has increased our ability to form potential models for how splicing kinetics may be regulated *in vivo.*

Highlights

Techniques to study splicing kinetics include metabolic labeling time courses, 3′ splice site-RNA polymerase II distance-based measurements, and microscopy-based methods.

The global median splicing time is of the order of minutes in many organisms, including yeast, *Drosophila*, and humans. Many instances of splicing events occurring on the order of seconds have also been observed.

Single intron excision kinetics are highly variable.

Many molecular features are associated with splicing kinetics, including intron length, the distance of an intron from its respective gene boundaries, and whether a splicing event is alternative or constitutive.

Deciphering when nascent RNA transcripts are spliced

Splicing, the joining of exons and removal of introns from **nascent RNA** (see Glossary), is an essential step in eukaryotic mRNA processing. Splicing regulation plays a critical role in many cellular processes, including, but not limited to, organism development [1,2], neuronal maturation [3,4], T cell development [5], and signaling pathways [5]. Furthermore, splicing defects have been implicated in many adverse health conditions, including cancer [5] and neurodegenerative disorders [6]. A burgeoning area of therapeutics seeks to target and correct mis-splicing events to prevent disease [7,8].

Many *in vitro* biochemical [9–11] and structural studies [12,13] have characterized splicing mechanisms in detail [14]. However, *in vitro* splicing studies often rely on purified splicing components or cellular lysates containing splicing factors and substrates at nonphysiological concentrations. Thus, these studies cannot recapitulate *in vivo* splicing dynamics, limiting what we can learn about how splicing timing influences different splicing outcomes. For instance, introns are transient *cis*-element landscapes; many contain splicing regulatory protein binding sites. As such, accurately measuring **splicing kinetics** *in vivo* allows us to study how the timing of intron removal affects the ability of splicing regulators to bind to nascent RNA and influence nearby splicing events [15] (Box 1). Moreover, many splicing events occur co-transcriptionally [16–20], linking splicing outcomes and dynamics with ongoing transcription *in vivo*. For example, multiple 3′ splice sites (SSs) may compete for the same 5′ SS, resulting in **alternative splicing** events. However, if the upstream 3′ SS is used in the splicing reaction with the 5′ SS before the downstream 3′ SS is transcribed, this source of kinetic competition is abolished. Therefore, measuring splicing kinetics *in vivo* has the potential to increase our understanding of how various splicing outcomes are reached.

Technical challenges previously prevented global and physiological measurements of splicing kinetics *in vivo*. Now, advances in the past few years, in particular long-read sequencing and metabolic labeling strategies, have made it possible to make increasingly accurate measurements

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Box 1. Splicing order and coordination relate to splicing kinetics

The relative order of intron removal within one transcript is directly influenced by splicing kinetics. If splicing happens more quickly than the synthesis of downstream introns, then splicing should predominantly occur in the transcription order. Conversely, out-of-order splicing presumes that upstream introns are excised after downstream introns have been synthesized. Estimates for genome-wide out-of-order splicing frequency between consecutive exon junctions pairs range between 12 and 54% in human cells [25,46,68,69]. Most studies have reported breakdowns close to 50%, similar to previous observations in *S. pombe* [34], while in-order splicing appears more frequent in *Drosophila* cells (65%), consistent with faster splicing kinetics in this species [25]. Interestingly, splicing order between adjacent exon junctions pairs is very consistent across mammalian species, cells, and tissue types [69].

Increasing evidence also suggests that there is coordination or coupling between the removal of proximal introns [10,15,25,31,68]. For example, neighboring introns are more likely to have the same excision status than more distal introns within single nascent transcripts in human cells [25]. Introns within the same gene are also more likely to have similar excision rates or levels in nascent RNA than expected by chance [52,62]. Certain introns in minigene assays could only be efficiently removed if their neighboring introns were also present [68], suggesting that the excision kinetics of each intron may be influenced by those of its neighbors and the surrounding *cis*-regulatory context. By contrast, excision of proximal or distal introns pairs appeared to be independent from one another in single-molecule RNA fluorescence in situ hybridization (FISH) analysis of a few genes [63]. Thus, the extent of splicing coordination may vary across the transcriptome.

Various observations have begun to elucidate the principles that regulate splicing coordination. *In vitro* experiments have shown that recruitment of U2 small nuclear ribonucleoprotein (snRNP) to a 3′ SS is enhanced by the presence of U1 snRNPs at both the upstream and downstream 5′ SS, demonstrating that synergistic spliceosome assembly may coordinate intron removal [10]. In *Drosophila*, deposition of the exon junction complex was proposed to contribute to excision coupling between two introns [70]. Further investigating the roles of the spliceosome and other RNA binding proteins in splicing coupling will be critical to building a mechanistic understanding of splicing coordination.

of splicing kinetics [21–25]. These quantitative measurements are an important step towards understanding the molecular features that shape splicing rates and elucidating how splicing kinetics contribute to isoform diversification.

Technical advances and approaches for quantifying splicing kinetics

One strategy for studying splicing kinetics is measuring the timing of splicing events through a time course. Establishing a starting time point for these measurements is difficult, as intron excision is ongoing and asynchronous among introns. To circumvent this issue, some studies induced the expression of specific genes [26]; an approach that cannot be generalized to the whole transcriptome. Other studies treated cells with a transcriptional inhibitor to synchronize the initiation of RNA synthesis upon inhibitor removal [27]; a valuable but nonphysiological approach. Therefore, to measure splicing kinetics physiologically on a genome-wide scale, new approaches were necessary. In the following section, we will review three methods that have been developed to study splicing kinetics: metabolic labeling-based time course measurements, 3′ SS-RNA polymerase II (Pol II) distance-based measurements, and microscopy-based measurements.

Metabolic labeling-based time course measurements

Splicing kinetics can be measured through time courses using metabolic labels, where a modified nucleotide is taken up by cells and incorporated into newly synthesized transcripts (Figure 1A). Labeled RNA are distinguished from background RNA, allowing for the identification of RNA synthesized during the labeling period. To determine splicing kinetics, splicing completion is measured as a function of time by quantifying the proportion of spliced RNA within the metabolically labeled RNA fraction across several sampling time points (Figure 2, top panel) [22–24]. Metabolic labeling enables genome-wide time course measurements while maintaining physiological transcription, decreasing the bias that might come from gene-targeted or transcriptional arrest-based measurements.

Careful experimental design and analysis are critical to calculating accurate splicing kinetic rates from metabolic labeling time course experiments. Currently, the most commonly used metabolic

Glossary

Alternative splicing: splicing event in which exons can be combined in more than one way, resulting in different splicing outcomes. Alternative splicing events include alternative 5′ SSs, alternative 3′ SSs, intron retention, exon skipping, and mutually exclusive exons.

Constitutive splicing: splicing event in which exons are always combined in the same way, resulting in a singular splicing outcome.

Early splicing event: rapid splicing event compared to global median estimates of splicing times, occurring on the order of seconds and when RNA Pol II is within a few hundred base pairs from the 3′ SS used in the splicing reaction.

Excision kinetics: speed at which introns are removed either globally or for individual introns. This term is intron-centric, in comparison to the term splicing kinetics, which refers to the speed at which two exons are combined through excision of an intron.

Nascent RNA: RNA undergoing synthesis and/or maturation. Nascent RNA is enriched for pre-mRNA and RNA undergoing processing events such as splicing. In this review, we consider both co-transcriptional and post-transcriptional RNA to have the capacity to be nascent while it is still being processed into its mature isoform.

Recursive splicing: segmental removal of an intron, whereby excised segments are flanked at one or both ends by an exon 0 nucleotides in length (i.e. the 3′ SS is adjacent to a 5′ SS), resulting in sequential splicing events. For example, upon splicing of a 0-nucleotide exon to an upstream exon, the 5′ SS is regenerated by the 0-nucleotide exon and can be used in subsequent splicing reactions with a downstream 3′ SS [65–67].

Splicing efficiency: comparative measure of how completely a splicing event occurs (i.e., what percentage of molecules undergo a splicing event), measured relative to other splicing events. Splicing events that have higher splicing efficiencies are often interpreted to have faster kinetics than those with lower splicing efficiencies, especially when analyzing nascent RNA actively undergoing splicing. However, when comparing across splicing events, a higher or lower splicing efficiency does not necessarily indicate faster or slower splicing kinetics respectively. For example, some exon junctions may be

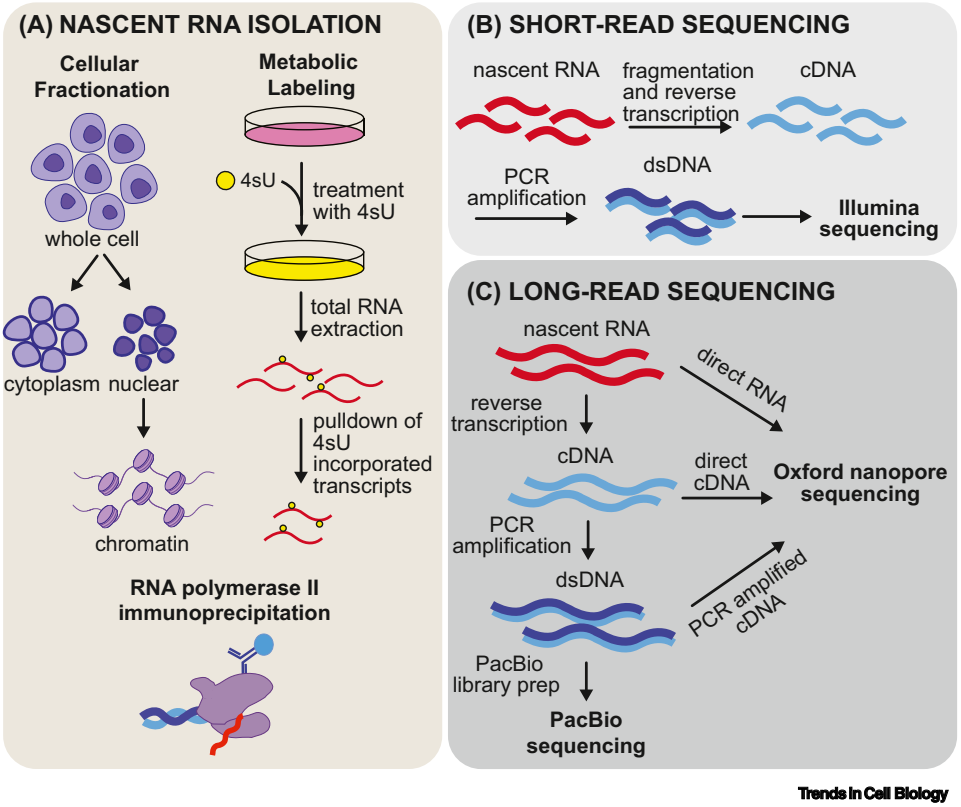

Figure 1. Schematic of protocols for studying splicing kinetics using short- and long-read sequencing approaches. (A) Different methods to isolate nascent RNA include cell fractionation to extract chromatin-associated RNA, 4sU-based metabolic labeling, and Pol II immunoprecipitation to isolate RNA undergoing synthesis. Nascent RNA can be sequenced using (B) short-read sequencing or (C) long-read sequencing, the latter of which can be conducted on RNA, cDNA, or PCR-amplified cDNA. Abbreviations: 4sU, 4-thiouridine; dsDNA, double-stranded DNA; Pol II, RNA polymerase II.

spliced quickly but may only be spliced in a small fraction of transcripts, leading to low calculated splicing efficiencies despite fast splicing kinetics.

Splicing kinetics: the speed at which splicing occurs either globally or for individual exon junctions. This speed can be measured in units of time or in nucleotides, by measuring the distance RNA Pol II travels from a 3′ SS prior to excision of the given intron. By measuring the RNA Pol II elongation speed, nucleotide measurements of splicing kinetics can be converted to time estimates.

label is 4-thiouridine [4sU, or 4-thiouracil (4tU) in yeast] [22–25,28,29]. To isolate metabolically labeled RNA, studies often biotinylate 4sU-labeled RNA and conduct a streptavidin pulldown [23–25]. While transcripts undergoing transcription will incorporate a metabolic label at their 3′ ends, pulldown of intact, labeled transcripts can also capture 5′ nonlabeled transcript segments, resulting in a bias towards 5′ reads in short-read sequencing libraries. To evenly sample reads across transcripts, RNA can be fragmented prior to pulldown [24]. Different analytical approaches have been developed for metabolic labeling data. Commonly, splicing rates are calculated from junction reads (i.e., intron–exon, exon–intron, or exon–exon reads) [23,24,28]. This approach is in contrast to estimating rates based on the proportion of intron-containing transcripts or intronic reads at different timepoints. While the latter method provides better coverage for lowly expressed genes, the former approach better estimates splicing rates from simulated data and avoids underestimating splicing rates due to reads from incompletely synthesized introns and degraded transcripts [23].

To measure splicing kinetics accurately using metabolic labeling, the concentration and labeling period used are important to consider. Studies have shown that 4sU can be used across a range of concentrations (40–100μM) for short periods of time (2–4 h) without significantly affecting splicing in cell culture [25,30]. At extended labeling times (24 h) or at high concentrations,

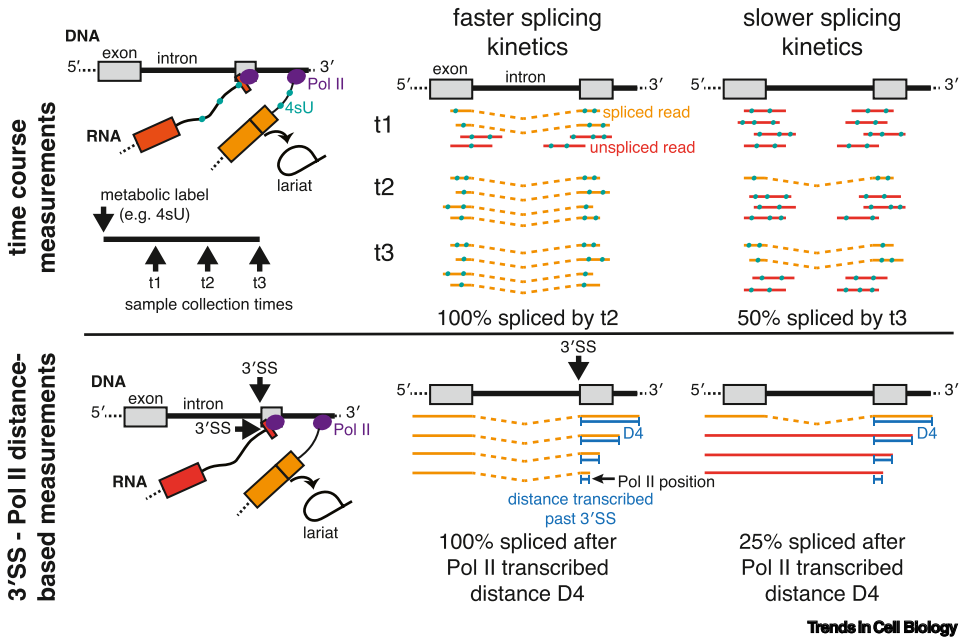

Figure 2. Schematics of the two most common sequencing-based approaches to study splicing kinetics. Top: time course measurements rely on labeling newly synthesized RNAs with a nucleotide analog, collecting samples at different timepoints, and measuring the fraction of spliced reads at each timepoint. Here, the nucleotide analog used is 4sU. Bottom: 3′ SS-Pol II distance-based measurements rely on computing the fraction of spliced reads as a function of the distance between the 3′ SS and the end of the read, used as a proxy for Pol II position. Schematics on the right show example reads as a function of time or distance between 3′ SS and Pol II position for introns with fast or slow excision kinetics. Abbreviations: 4sU, 4-thiouridine; Pol II, RNA polymerase II; SS, splice site.

4sU affects splicing profiles [30], potentially by exhibiting cellular toxicity. 4sU has also been hypothesized to bias splicing profiles by decreasing the binding of small nuclear RNAs (snRNAs) to U-rich intronic elements, thereby leading to underestimations of splicing rates [31], although this concern has not been directly tested. Overall, when used for a short amount of time and at an appropriate concentration, metabolic labels allow for the measurement of splicing times genome-wide and at near-physiological conditions.

3′ SS-Pol II distance-based estimations

Co-transcriptional splicing kinetics can be measured through 3′ SS-Pol II distance-based measurements, whereby the excision status of an intron is compared with the distance between its 3′ SS and the downstream Pol II that synthesized it (Figure 2, bottom panel) [32]. 3′ SS-Pol II distances serve as a proxy for the time that elapses between the synthesis and excision of introns [25,31]. Sequencing nascent RNA simultaneously reveals the 3′ end, which represents the position of Pol II, and the excision status of all introns within the sequencing read. With short-read sequencing, distance measurements between Pol II and SSs are limited to several hundred nucleotides, which are insufficient to capture many splicing events (Figure 1B). By contrast, nanopore direct RNA, nanopore cDNA, or PacBio long-read sequencing technologies [33] sequence hundreds to thousands of nucleotides past a 3′ SS of interest within a single read [21,25,31,34] (Figure 1C).

3′ SS-Pol II distance-based measurements have unique advantages when compared with other techniques to measure splicing kinetics. Notably, as distance-based estimates are measured in nucleotides between Pol II and the 3′ SS, they are able to capture fast splicing events at high

resolution. For splicing events occurring within several to tens of seconds after transcription of the 3′ SS, these distance-based measurements may be more accurate than time course sequencing experiments, which typically are collected over the range of minutes [22–24]. Additionally, as distance-based measurements capture Pol II position as well as splicing activity, they allow for insights into splicing regulation *in vivo*, such as the dependence of different splicing events on the greater transcription/excision status of the pre-mRNA (e.g., exon and intron definition models that have been reviewed elsewhere [35] (Box 2).

Depending on the sequencing technology and protocols used, 3′ SS-Pol II distance-based measurements have several limitations. First, long-read sequencing approaches, especially direct RNA sequencing, have suffered from low depth and throughput. Long-read sequencing studies performed at low sequencing depths have enabled global estimates of splicing kinetics, but have been more limited with respect to the analysis of excision events of individual introns [33]. Conversely, for long-read approaches that do not sequence RNA directly, strand-switching reverse transcription and/or PCR amplification are necessary for library preparation. These steps preferentially capture shorter, spliced RNAs, which can lead to the overestimation of splicing events and underestimation of splicing rates [36]. Second, 3′ SS-Pol II distance-based analyses often use an average Pol II elongation rate [32] to convert base pair estimates to a time estimate, even though Pol II pauses throughout the gene body and travels at different rates across exons and introns [37–41]. Therefore, assuming one Pol II elongation rate may lead to inaccurate conversions between base pair and time-based splicing kinetics calculations, although it can provide order of magnitude estimates. Finally, many distance-based measurements rely on the isolation of nascent RNA through cell fractionation for chromatin-associated RNA. This RNA fraction may be contaminated with mature and degraded RNAs. To reduce this background, nanopore analysis of co-transcriptional processing (nano-COP) combines 4sU metabolic labeling and isolation of labeled transcripts with cell fractionation to improve enrichment for nascent RNA [25,36].

Microscopy-based approaches

Live cell imaging is an orthogonal method to sequencing-based approaches to measure splicing timing. Specific introns can be labeled for imaging through the insertion of reporter sequences,

Box 2. Exon and intron definition models of splicing

The first step in intron excision is the recognition of the 5′ and 3′ SSs by the splicing machinery to define the intron to be removed. In the 1990s, two models, described in depth in reviews elsewhere [35,71], were put forth to explain how SSs are recognized depending on the gene structure. The intron definition model posits that the 5′ and 3′ SSs are recognized and paired across an intron. This model is thought to predominate when short introns are separated by longer exons (e.g., in yeast and for most introns in *Drosophila*). By contrast, the exon definition model hypothesizes that the 5′ and 3′ SSs are recognized and paired across an exon, followed by a rearrangement to pair the 3′ SS with the 5′ SS of the upstream intron. This model is thought to prevail when long introns are separated by short exons, such as in mammalian genomes. Evidence for these models came primarily from genetic mutations, either introduced in model genes or identified in human patients [71].

Recent studies are starting to paint a more complex picture, suggesting that both models may be used in mammalian cells. For example, mutagenesis experiments of two human genes suggest that even short introns (<100 nucleotides) can be removed through exon definition [15,72]. By contrast, distance-based measurements have revealed early splicing events in mammalian cells, in which introns are sometimes removed before the downstream exon is fully synthesized [21,46]. In these cases, which were observed for all intron sizes [46], the intron is excised independently of downstream 5′ SSs and is therefore removed through intron definition, suggesting that this mode of intron removal occurs more frequently and for longer introns than previously thought. Nevertheless, most splicing events in these studies occurred after synthesis of the downstream 5′ SS and thus do not exclude exon definition [21,46]. *In vitro* demonstration of synergistic spliceosome assembly across exons is also consistent with exon definition [10]. It may be that while splicing can occur through intron definition in humans, its efficiency and/or kinetics are enhanced by exon definition, which would enable the coordination of neighboring splicing reactions (see Box 1 in the main text) [25]. Finally, the frequent occurrence of recursive splicing (see Box 4 in the main text) suggests that intron excision may not always occur through either of the two models, although intron or exon definition may contribute to the recognition of exon boundaries.

such as MS2 or PP7 stem loops, which are recognized by fluorescent protein-tagged versions of their respective coat proteins. Splicing kinetics are then measured through tracking the appearance (synthesis) and disappearance (splicing) of intronic fluorescent reporter sequences [42,43], allowing for real-time measurements on the scale of seconds [42]. Because reporters can be inserted anywhere within a gene, **excision kinetics** can be measured for any intron of choice, and various introns within the same gene can be analyzed, regardless of the distance between them.

Several drawbacks complicate the study of splicing kinetics through live cell imaging. The insertion of exogenous reporter genes and sequences may disrupt RNA processing [44], resulting in altered measurements. Likewise, although intronic signal disappearance is often interpreted as a splicing event, transcript decay and premature termination cannot be ruled out based on microscopy data alone [42]. Live cell imaging has historically relied on the preselection of a small number of genes, resulting in a potential selection bias [32,45]. To address this concern, recent high throughput methods have used random intronic integration of reporter sequences to measure excision kinetics in hundreds of endogenous introns [42].

Estimates of splicing times

New technologies to study splicing kinetics now enable more accurate measurements than previously possible, providing novel insights into splicing dynamics. These measurements have revealed a wide variability in observed splicing times [22,42,46]. More accurate measurements have also led to observations of **early splicing events**, which can occur almost immediately after an intron is synthesized [21,46,47].

Splicing kinetics are variable across introns

Recent studies of splicing kinetics have highlighted their variability, which can stretch across orders of magnitude. For instance, fission yeast *Schizosaccharomyces pombe* median splicing times and budding yeast *Saccharomyces cerevisiae* median splicing half-lives have been estimated to be anywhere from 37 to 135 s, respectively (Table 1), with a greater than 30-fold variation in splicing half-lives observed across the *S. cerevisiae* transcriptome [22,29]. In human cells, both microscopy-based and direct RNA sequencing nano-COP analyses estimate median splicing times to be on the order of minutes, ranging from 1 to 8 min [24,25,42] (Table 1). Global splicing times can also vary by up to five orders of magnitude, as the distance between Pol II and the 3′ SS for splicing events has been measured to range from 10 to 100,000 nucleotides [46]. Assuming an average Pol II elongation rate of approximately 2 kb/min [38,48], these results correspond to a range of splicing times spanning from 0.3 s to 50 min. To measure this wide range of splicing times, nascent lariat intermediates undergoing splicing were captured by co-transcriptional lariat sequencing (CoLa-seq), and the distance between the 3′ SS and Pol II was measured. Although short-read sequencing was used, large 3′ SS-Pol II distances could be recovered from reads where downstream excision events of long introns allowed for the inference that Pol II had already traversed the length of the excised intron [46].

Splicing times are also variable across single exon junctions. Live imaging studies have measured excision times for single introns ranging from minutes to an hour [42]. CoLa-seq has similarly found evidence for significant variation in excision timing at the single-intron level [46]. Going forward, further studies are needed to more deeply characterize the variability of individual intron excision times genome-wide.

Capturing early splicing events

Within the large variation of splicing times reported, 3′ SS-Pol II distance-based measurements have captured early splicing events at nucleotide resolution. In budding yeast, many splicing events have

Table 1. Summary of different splicing kinetics determined experimentally

Model organism	Technique name	Nascent RNA isolation and/or labeling method	Reverse transcription and/or amplification	Sequencing or signal readout method	Splicing kinetics estimation (3′ SS-Pol II distance)	Median splicing time estimation	Approx. no. of genes analyzed	Refs
Budding yeast *S. cerevisiae*	Single molecule intron tracking (SMIT)	Cell fractionation for chromatin-associated RNA	Targeted reverse transcription and PCR amplification	Short-read sequencing (Illumina HiSeq)	50% of introns excised after Pol II travels 45 nucleotides from the 3′ SS	1.35 s[a]	87 genes	Oesterreich *et al.* 2016 [47]
Human K562 cells	Co-transcriptional lariat sequencing (CoLa-seq)	Cell fractionation for chromatin-associated RNA, degrade linear RNA and capture RNA containing splicing lariats	Reverse transcription and PCR amplification	Short-read sequencing (Illumina HiSeq)	50% of introns excised after Pol II travels 150 nucleotides from the 3′ SS	4.5 s[a]	1000s of genes	Zeng *et al.* 2022 [46]
Murine MEL cells	Not applicable (NA)	Cell fractionation for chromatin-associated RNA	Reverse transcription with strand-switching and PCR amplification	cDNA long-read sequencing (PacBio)	50% of introns excised after Pol II travels 154 nucleotides from the 3′ SS, when considering the cumulative distribution of spliced reads only	4.6 s[a]	1000s of genes	Reimer *et al.* 2021 [31]
Human HeLa cells	Polymerase intact nascent transcript technology (POINT-nano)	Chromatin isolation, DNase digestion, immunoprecipitation of the intact Pol II and nascent RNA complex	Reverse transcription with strand-switching	Direct cDNA long-read sequencing (Oxford Nanopore)	50% of introns excised after Pol II travels ~2 kb past the 5′ SS of the downstream intron	~1 min[a,b]	1000s of genes	Sousa-Luís *et al.* 2021 [21]
Drosophila S2 cells	Nano-COP	4sU labeling, cell fractionation for chromatin-associated RNA and 4sU pulldown	NA	Direct RNA sequencing (Oxford Nanopore)	50% of introns excised after Pol II travels ~1750 nucleotides from the 3′ SS	1.75 min[a]	1000s of genes	Drexler *et al.* 2020 [25]
Human K562 cells	Nano-COP	4sU labeling, cell fractionation for chromatin-associated RNA and 4sU pulldown	NA	Direct RNA sequencing (Oxford Nanopore)	<20% of introns excised after Pol II travels ~2 kb past the 3′ SS	>1 min[a]	>10 000 genes	Drexler *et al.* 2020 [25]
Drosophila S2 cells	NA	4sU metabolic labeling over a time course and labeled RNA pulldown	Reverse transcription and PCR amplification	Short read sequencing (Illumina HiSeq)	NA	2 min	1000s of genes	Pai *et al.* 2017 [23]
Budding yeast *S. cerevisiae*	Multiplexed primer-extension sequencing (MPE-seq) [78,79]	4tU metabolic labeling over a time course and labeled RNA pulldown	MPE-seq library preparation	MPE-seq/ Short-read sequencing (Illumina NextSeq)	NA	2.25 min	100s of genes	Gildea *et al.* 2022 [22]
Human K562 cells	Transient transcriptome sequencing (TT-seq)	4sU metabolic labeling over a time course and labeled RNA	Reverse transcription and PCR amplification	Short read sequencing (Illumina HiSeq)	NA	7.2 min	1000s of genes	Wachutka *et al.* 2019 [24]

Table 1. (continued)

Model organism	Technique name	Nascent RNA isolation and/or labeling method	Reverse transcription and/or amplification	Sequencing or signal readout method	Splicing kinetics estimation (3′ SS-Pol II distance)	Median splicing time estimation	Approx. no. of genes analyzed	Refs
		pulldown						
Human HBEC cells	NA	NA	NA	NA	NA	8.2 min[c]	10 genes	Wan *et al.* 2021 [42]
Lipopolysaccharide stimulated mouse immune dendritic cells	NA	Time course following lipopolysaccharide stimulation sampling nascent and total RNA; 10 min 4sU metabolic labeling pulses precede sampled time points; labeled RNA pulldown for nascent RNA samples	Reverse transcription and PCR amplification	Short read sequencing (Illumina HiSeq)	NA	14 min	>1000 genes	Rabani *et al.* 2014 [50]
Arabidopsis seedlings	NA	Cell fractionation for chromatin-associated RNA	Reverse transcription with strand-switching and PCR amplification	cDNA sequencing (Oxford Nanopore and PacBio)	<50% of introns excised when Pol II travels 1.1 kb past the 3′ SS	NA	>10 000 genes	Jia *et al.* 2020 [64]

[a]Indicates an approximate calculated time estimation based on 3′ SS-Pol II distance-based measurements, assuming an elongation rate of 2 kb/min in mammalian and yeast cells [38,48,75] and an elongation rate of 1 kb/min in *Drosophila* S2 cells [76,77]. More accurate time conversions from 3′ SS-Pol II distance-based measurements will be dependent on elongation rate measurements specific to cell type, condition, and local genomic context.
[b]This measurement does not take into account the distance elongating through the 3′ SS proximal, downstream exon and excludes immediate splicing events occurring when Pol II is in the 3′ SS proximal, downstream exon.
[c]The median splicing time estimation was determined from modeling. Model inputs included live cell imaging dynamics, smFISH, and RNA half-life measurements. Live cell imaging dynamics were captured through MS2 or PP7 stem loop integration into introns of endogenous genes. RNA half-lives were measured through a 4sU pulse-chase labeling strategy and probe-based qPCR assay.

been measured to occur within 45 nucleotides of the 3′ SS [34,47]. In human cell lines too, splicing events have been observed as early as when Pol II is 10 to 15 nucleotides downstream of the 3′ SS [21,46]. In line with observations of early splicing, several studies have fit the probability of intron excision over time to an exponential decay model [42,46], consistent with a single rate limiting step in intron removal. Critically, the expectations of this model are that early splicing events around $t = 0$ have the highest probability of occurring, while the median excision time is higher than the most commonly observed times and is set by the distribution decay rate. Due to the long tail of the exponential distribution, this simple model is consistent with reports that splicing happens both immediately upon 3′ SS synthesis and after Pol II has transcribed further [21].

Although early splicing events are expected for exponential decay models of intron excision [46], studies have concluded considerably differing estimates of their frequencies, which in some cases were high. Thus, the extent of early splicing events genome-wide remains an active area of investigation. Differing estimates may be due to differences in data analysis approaches

(Box 3) and technical reasons. A majority of the studies reporting widespread early splicing events have relied on cDNA synthesis with template strand-switching, and/or PCR amplification of cDNA prior to short- or long-read sequencing [21,31,34,46,47]. As these library preparation methods bias towards shorter, spliced reads [36], they are likely to oversample early splicing events. Furthermore, it has been speculated that use of 4sU biotinylation and streptavidin-based enrichment of long pre-mRNAs by the nano-COP approach could bias towards unspliced species, which are longer than spliced species and therefore contain more incorporated 4sU, potentially rendering their pulldown more likely [31]. This bias would favor the capture of late splicing events, although analyses comparing splicing kinetics with or without 4sU-based enrichment for nascent RNA have shown similar results [36]. Moreover, nano-COP estimates of splicing times are aligned with estimates from microscopy-based approaches [42,49] and metabolic labeling-based time course measurements [24,50]. These caveats highlight the importance of confirming splicing kinetics measurements using a variety of complementary techniques.

Features that influence splicing kinetics

Intronic properties influence splicing kinetics

One of the most pressing questions regarding splicing kinetics is determining what features influence the rate of splicing. Many features have been associated with splicing kinetics and efficiency: GC content of introns and their flanking exons [28], gene expression levels [51,52], SS strength [22,23,28,53,54], RNA binding protein binding patterns [24], nascent RNA structure across introns and at SSs [54], the coding versus noncoding status of RNAs [24,28,29,55], and exon length [46]. We focus here on the discussion of two intronic properties that substantially influence splicing kinetics: intron length and the distance between an intron and its respective gene boundaries.

Many studies have found that intron length influences splicing kinetics [23–25,46,53]. Surprisingly, although positive correlations between overall gene length and **splicing efficiency** have been reported [52], it does not appear that a simple linear relationship exists between intron length and the time it takes for an intron to be excised. In *Drosophila*, studies have shown that introns with a length of 60–70 nucleotides are excised faster on average than shorter or

Box 3. Methods to analyze 3′ SS-Pol II distances

No standard approach yet exists for calculating splicing kinetics from 3′ SS-Pol II distances in long-read sequencing data. On the individual gene level, looking at raw reads reveals fast and slow splicing events (Figure IA). However, when measuring 3′ SS-Pol II distances globally, more sophisticated analyses are necessary. Research studies employing various analyses have reached different conclusions regarding global splicing kinetics in mammalian cells [25,31], underscoring the importance of understanding the limitations of different approaches.

One approach to measuring 3′ SS-Pol II distances globally is to plot the cumulative fraction of splicing observations (Figure IB). Using this approach, up to 75% of 3′-proximal splice junctions have been reported to be spliced within a Pol II distance of 300 nucleotides from the 3′ SS [31]. Notably, this approach discards unspliced reads. As intron retention can lead to an overestimation of splicing times, this approach avoids this bias. However, by only analyzing splice junctions, this analysis assumes co-transcriptional splicing events will occur within the 3′ SS-Pol II distances covered by long-read sequencing read lengths. It also assumes that any unspliced reads are unlikely to ever be spliced, which is at odds with research suggesting that splicing rate and yield are distinct and do not correlate with each other [24].

Distances between the 3′ SS and Pol II have also been measured against the percent of spliced reads to estimate global splicing kinetics. When analyzing both spliced and unspliced exon junctions using this approach, less than 20% of introns are excised when Pol II is within 1.5 kb of the 3′ SS in human cells [25] (Figure IC). This analysis corresponds with observations that splicing can occur immediately after 3′ SS synthesis, yet is ongoing when Pol II is >2 kb downstream from the 3′ SS, with a significant RNA fraction still unspliced [21]. Therefore, by including unspliced and spliced reads in the splicing kinetics estimation, this analysis considers reads that are often unspliced within the 3′ SS-Pol II distances sequenced, but may be spliced later. Inherently, this approach makes assumptions about if a given intron will be eventually excised or retained. To avoid the inclusion of retained introns, the analysis can be limited to introns that are highly excised in short-read sequencing data of total RNA collected under the same conditions and in the same cell line [25,36].

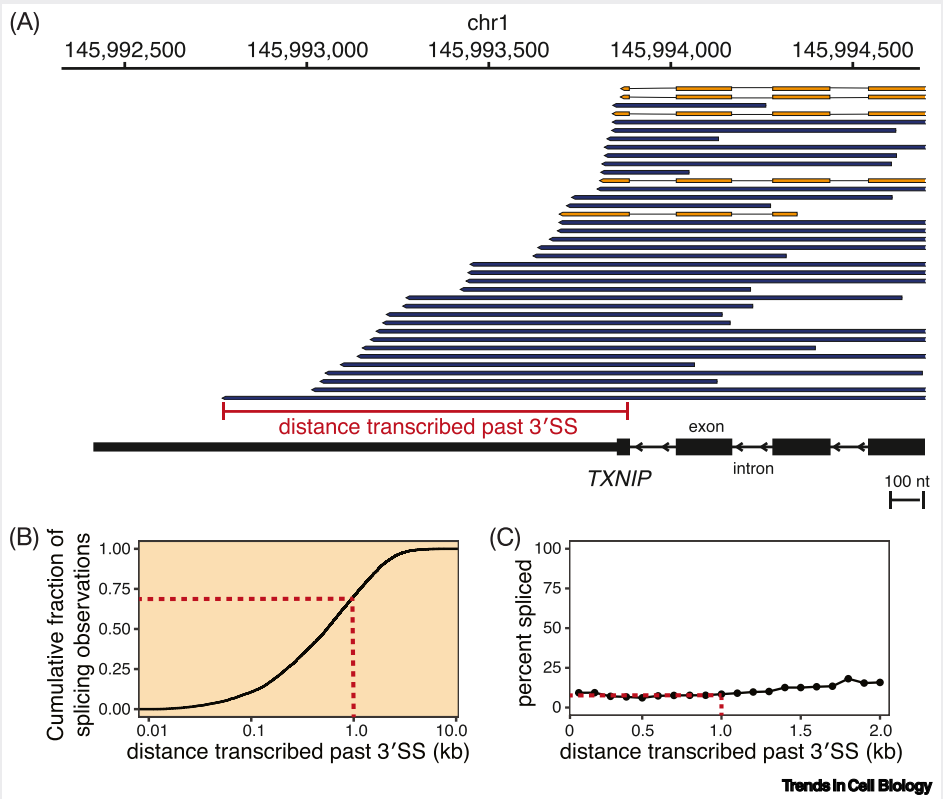

Figure I. Splicing kinetics estimations using the distance between Pol II and the 3′ SS can be quantified using different approaches. (A) Long-read direct RNA sequencing reads from K562 cells [25] overlapping with the last intron of *TXNIP*. The 3′ end of the read represents the Pol II position. Instances of fast splicing are shown in yellow, while other reads display slower splicing. (B,C) Different methods to analyze 3′ SS-Pol II distance-based measurements result in different splicing kinetics estimations. (B) Only reads in which the introns of interest are excised are used to calculate the cumulative fraction of splicing observations as a function of the distance transcribed by Pol II past the 3′ SS. (C) All reads mapping to introns of interest are used to calculate the percent of reads that are spliced as a function of the distance transcribed by Pol II past the 3′ SS. The red dashed lines indicate the *y*-axis value equivalent to 1 kb for each plot. Data from introns in protein-coding genes in human K562 cells from Drexler *et al.* 2020 [25] were used to make the two plots [(B) n =6651 read/intron pairs, (C) n =72,267 read/intron pairs]. Prior to plotting, data was processed as in [25]. Abbreviations: Pol II, RNA polymerase II; SS, splice site.

longer introns, suggesting that an optimal intron length may exist [23]. Similarly, in *Drosophila* embryos, excision occurs most efficiently for introns that are 55–100 nucleotides, as well as introns >1 kb [53]. Therefore, it may be that in *Drosophila* embryos, several strategies exist to excise introns of different lengths in an efficient manner. For example, introns that strongly favor an intron or exon definition mode of excision (Box 2) may be more rapidly removed [23]. In human cells, it has been observed that for introns longer than one kilobase, in-order excision of the upstream intron becomes increasingly likely, highlighting how intronic length may affect kinetic competition between neighboring introns [46]. By contrast, splicing kinetics in *S. pombe* do not correlate with intron length and negatively correlate with transcription unit length [29]. It is possible this discrepancy reflects the overall shorter gene and intron lengths in *S. pombe* [29], where it may not have been necessary to evolve efficient strategies to excise long introns or introns in transcripts synthesized from long genes. Future studies may aim to better understand what underlying mechanisms account for the complicated relationship between intron length and splicing kinetics, including studying factors such as RNA secondary structure

and **recursive splicing**, the latter of which has previously been shown to vary with intron length [42] and may influence an intron's excision kinetics (Box 4).

A second feature repeatedly noted for its role in influencing splicing kinetics and efficiency is the distance of an intron from its respective transcription start and end sites [21,28,34,52]. In fission yeast, it has been reported that internal introns demonstrate significantly higher excision efficiencies than introns in single-intron genes, first introns, or last introns [34]. In human data, splicing speed correlates positively with intronic distance from transcription start and end sites [28], and last introns are often excised post-transcriptionally [25]. Thus, internal introns tend to be excised faster than introns at gene boundaries. However, it is unclear why internal introns demonstrate more efficient excision. Both first and last introns represent gene regions where intricate regulation exists among capping, splicing, and termination processes, which may prevent efficient excision [52].

Splicing events with distinct splicing kinetics

Alternative splicing events generally demonstrate slower splicing kinetics than **constitutive splicing** events. In *Drosophila* and human cells, excision kinetics for introns neighboring constitutively spliced exons are faster than those neighboring alternatively spliced exons [21,25,46]. In mammalian cells, introns flanking alternatively spliced exons are associated with weaker SSs and are more likely to be excised post-transcriptionally than those flanking constitutively spliced exons, consistent with their relatively slower excision kinetics [56–59]. These observations lead to the intriguing possibility that slower splicing kinetics allow for more precise discrimination between competing alternative SSs. However, an equally plausible explanation is that the higher regulation of splicing at alternative SSs results in slower splicing kinetics. Differing properties associated with constitutive and alternative splicing events, such as the generally longer length of introns adjacent to alternatively spliced exons [23,60], could also contribute to differences in splicing kinetics between the different classes of splicing events.

Notably, intron retention and detention constitute a special type of regulated alternative splicing that must be carefully considered in the context of splicing kinetics. Retained introns are never excised and are included in cytoplasmic mRNAs, while detained introns exist long-term in nuclear transcripts until they are eventually excised or their transcripts are degraded [61]. Thus, these intron types may appear to have very long excision kinetics, but additional validation may be required to determine whether they are ever excised.

Box 4. Splicing kinetics in light of recursive splicing

Recent evidence has uncovered many more instances of recursive splicing than have previously been appreciated. Recursive splicing was only discovered to occur in human cells within the past decade [73]. Yet, recent experiments that enrich for nascent RNA have annotated thousands of recursive splice sites within the human genome, leading to estimates that introns from up to 30% of expressed genes may undergo recursive splicing [42]. These findings are complemented by past studies that have remarked upon the high numbers of splicing events that map to nonannotated donor and acceptor sites [24], as well as studies that have noted specific examples of potential co-transcriptional recursive splicing events in yeast and murine samples [31,74].

If recursive splicing is a common strategy for intron removal, it may add additional considerations to the quantification of splicing kinetics. When recursive splicing occurs, the splicing kinetics for the final excised intron represent an amalgamation of many smaller splicing events, each with their own potentially variable splicing kinetics. Studies have also remarked that, in human cells, lariat lengths do not scale proportionally with the full lengths of their respective introns, and that full length lariats are more likely to be observed for shorter (<1.5 kb) introns than longer introns [42]. This observation implies that shorter introns are less likely to undergo recursive splicing than relatively longer introns, and leads to the compelling question of how recursive splicing events might contribute to the relationships noted between intron length and splicing kinetics [46]. Finally, it will be informative to determine if recursive splicing kinetics change relative to their proximity to exonic boundaries, in order to better understand if and how recursive splicing kinetics are modulated across introns.

Concluding remarks

New insights into splicing kinetics have been uncovered using metabolic labeling [22–24], 3′ SS-Pol II distance-based measurements [25,31,34], and microscopy-based methods [42]. A notable achievement of 3′ SS-Pol II distance-based measurements has been to capture splicing kinetics at base pair resolution, allowing for the observation of early splicing events [21,34,47]. By quantifying splicing kinetics using various methods, it has become apparent that most splicing events do not immediately occur following intron synthesis. The median excision time for introns in human cells is on the order of several minutes [24,25,42,62], and in yeast, median splicing time estimates range from 30 s to several minutes[22,29]. The broad range of splicing times observed [22,46] highlights the overall variability in splicing kinetics, both globally and at the single intron level [42].

Moving forward, it will be critical to broaden the scope of splicing events measured and the contexts in which splicing kinetics are measured. Although 3′ SS-Pol II distance-based measurements capture the splicing kinetics of co-transcriptional events, many splicing events take place post-transcriptionally [15,63], and future studies should focus on quantifying the splicing kinetics of these post-transcriptional events. In addition, it will be crucial to continue studying features that shape splicing efficiencies. Mechanistic insights will increase our understanding of why specific features, such as the distance of an intron from its respective gene boundaries [21,28,34,52,53], are associated with different splicing efficiencies (see Outstanding questions). Finally, new studies have recently quantified if and how splicing kinetics change throughout important biological processes, including *Drosophila* development [53], signaling in response to environmental cues in plants [64], and mammalian erythropoiesis [31]. Continued studies aiming to understand how splicing kinetics influence biological processes will be critical to understanding the regulatory potential of splicing dynamics.

Outstanding questions

How widespread are early splicing events in cells? What techniques and analyses are most likely to minimize biases towards early splicing events when measuring them?

How are long introns defined and excised? If long introns undergo recursive excision, how different are splicing kinetics at canonical SSs compared with intron-internal, recursive SSs?

Why is there a nonlinear relationship between splicing kinetics and intron length? If there is an optimal intron length for efficient splicing within a given cell type, what determinants shape that optimal length?

Does RNA structure contribute strongly to splicing kinetics? If so, how?

Why are internal introns more efficiently excised than first or last introns?

Why do alternative splicing events demonstrate slower splicing kinetics than constitutive splicing events? Are the slower kinetics a cause or consequence of alternative splicing?

What are the fates of transcripts that are not co-transcriptionally spliced? How often are these transcripts post-transcriptionally spliced and how often are they degraded?

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Declaration of interests

No interests are declared.

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